debtkit

Debt sustainability analysis in R

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The question

How should we run the standard IMF and EC debt sustainability analysis workflow in a single reproducible R interface?

Twelve functions, one uniform schema, deterministic to stochastic in five lines.

Why it matters

- **Finance ministries** run annual debt sustainability analysis under IMF and EC templates
- **Central banks** assess fiscal risk in their financial stability reports
- **IMF Article IV missions** apply the Sovereign Risk and DSA Framework to every member
- **Sovereign-credit analysts** price default risk off debt-to-GDP trajectories

What is already out there

- **`vars`, `quantmod`**: time-series building blocks, no DSA templates[1]
- **IMF and EC DSA spreadsheets**: authoritative but closed, manual, hard to version
- **Bespoke ministry code**: reinvented on every project, inconsistent methodology
- **No R package implements the standard toolkit**

The gap: **no open-source package covers the full DSA workflow end to end**.

[1] Pfaff (2008), *Analysis of Integrated and Cointegrated Time Series with R*, Springer; Ryan & Ulrich (2024), *quantmod*, R package.

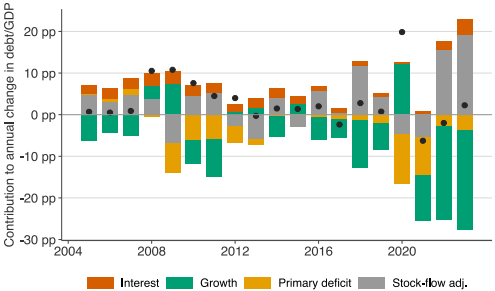
What debtkit offers

1. **Coverage**: 12 functions spanning projections, decomposition, Bohn tests, fan charts, IMF stress tests, heat maps, GFN, EC S1/S2 gaps
2. **Interface**: uniform `dk_` prefix, data-frame-in, S3-object-out, with `print()` and `plot()` methods
3. **Provenance**: pure computation, no data-vendor dependency, over 130 unit tests, on CRAN

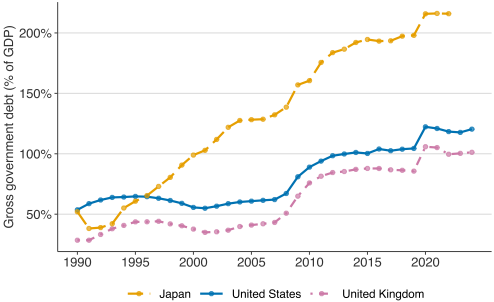
Methods follow Blanchard (1990), Bohn (1998), Celasun-Debrun-Ostry (2006), IMF (2022), and EC (2024)[2].

[2] Bohn (1998), *The Behaviour of US Public Debt and Deficits*, QJE 113(3); IMF (2022), *Staff Guidance Note on the Sovereign Risk and DSA Framework*; EC (2024), *Fiscal Sustainability Report 2024*.

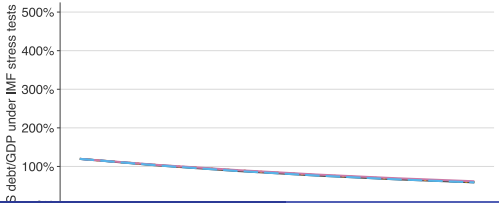
Breadth: one interface, four diagnostics

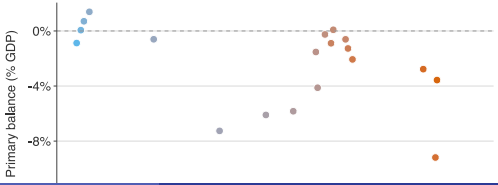


Historical decomposition: US 2004 to 2023



Cross-country trajectories: US, UK, Japan





Central formulas

Debt dynamics equation (Blanchard 1990):

$$d_{t+1} = \frac{1 + r_{t+1}}{1 + g_{t+1}} d_t - pb_{t+1} + sfa_{t+1} \tag{1}$$

Debt-stabilising primary balance:

$$pb^* = \frac{r - g}{1 + g} d \tag{2}$$

Bohn (1998) fiscal reaction test (sufficient condition: $\beta > 0$):

$$pb_t = \alpha + \beta d_{t-1} + \gamma' X_t + \varepsilon_t \tag{3}$$

Package at a glance

Function families:

- **Projection**: `dk_project()`, `dk_rg()`, `dk_compare()`
- **Decomposition**: `dk_decompose()`
- **Reaction function**: `dk_bohn_test()`
- **Stochastic**: `dk_estimate_shocks()`, `dk_fan_chart()`
- **Stress tests**: `dk_stress_test()`
- **Diagnostics**: `dk_heat_map()`, `dk_gfn()`, `dk_sustainability_gap()`
- **Helpers**: `dk_sample_data()`

Deps: `cli`, `grDevices`, `graphics`, `stats`. R $\geq$ 4.1.0.

Uniform interface

Every function accepts:
`debt`, `interest_rate`, `gdp_growth`, `primary_balance`
and returns an S3 object with `print()` and `plot()` methods.

Minimal working example

```r
library(debtkit)

# Five-line canonical DSA workflow
d      <- dk_sample_data()
decomp <- dk_decompose(d$debt, d$interest_rate, d$gdp_growth,
                       d$primary_balance, years = d$years)
shocks <- dk_estimate_shocks(d$gdp_growth, d$interest_rate,
                             d$primary_balance)
fan    <- dk_fan_chart(tail(d$debt, 1), 0.04, 0.03, -0.01,
                       shocks = shocks, n_sim = 2000, horizon = 10)
stress <- dk_stress_test(tail(d$debt, 1), 0.04, 0.03, -0.01)
```

Five lines take an analyst from raw fiscal series to the full IMF DSA template: decomposition, stochastic fan, and the six standardised stress tests.

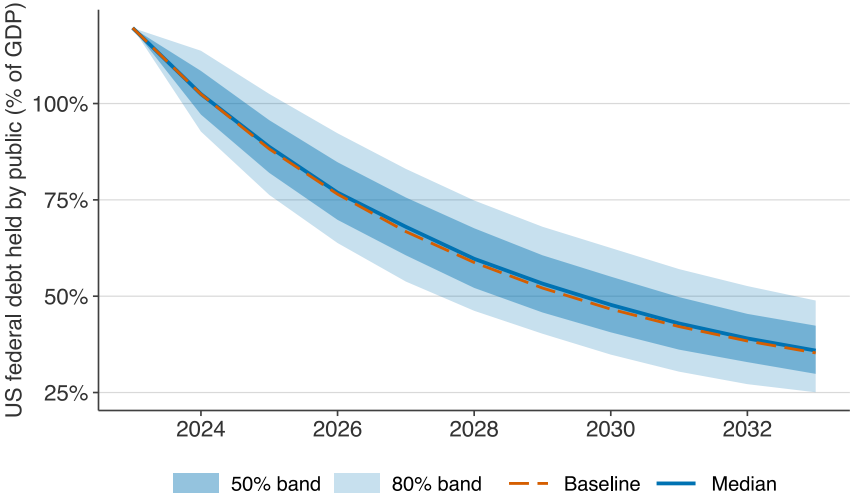
US federal debt dynamics after the pandemic

Data. FRED `GFDEGDQ188S` (US federal debt held by the public), `FYFSGDA188S` (federal balance), `IRLTLT01USM156N` (10-year yield), `A091RC1Q027SBEA` (GDP). Annual panel 2004 to 2023.

Question. *Is the US debt ratio, at 100 per cent of GDP, on a sustainable trajectory through 2033?*

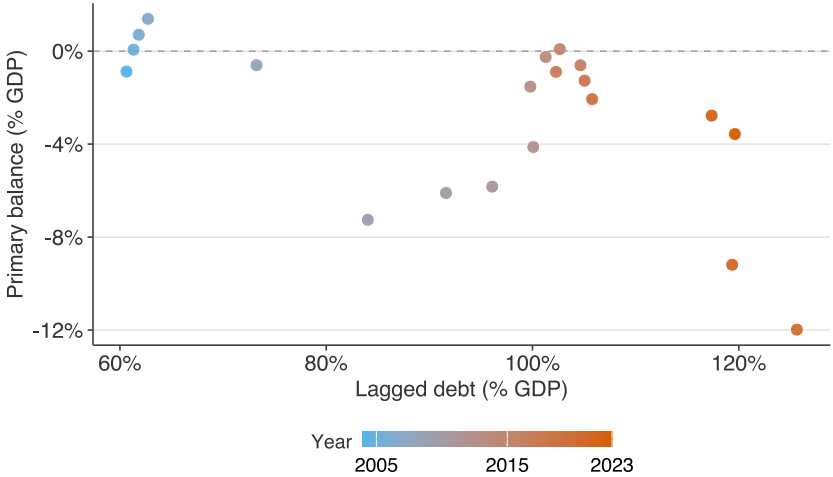
Why this case. The most-watched fiscal question in US policy since 2020[3]. Debt jumped from 79 per cent in 2019 to 100 per cent in 2021. Reproducible with the debtkit pipeline in under 20 lines.

[3] Blanchard (2022), *Fiscal Policy under Low Interest Rates*, MIT Press; Furman & Summers (2023), *A Reconsideration of Fiscal Policy in the Era of Low Interest Rates*, Brookings.



Ten-year stochastic fan chart for US federal debt held by the public, 2023 to 2033, 2000 Monte Carlo simulations. Inner band is the 50 per cent interval (25th to 75th percentile); outer band the 80 per cent interval. Solid blue line is the simulation median; red dashed line the deterministic baseline holding macro conditions fixed at 2023 values. Shocks from first-order VAR on 2004 to 2023 residuals. Source: FRED.

Bohn regression: does the primary balance respond to debt?



Bohn (1998) fiscal reaction function for the US 2004 to 2023. Points are annual observations, coloured by year. Red

What debtkit does not yet do

- **Annual frequency only**: IMF and EC templates are annual; quarterly DSA is possible but not standardised
- **First-order VAR or bootstrap shocks**: no regime-switching, time-varying-parameter, or stochastic-volatility models
- **No ageing-cost series bundled**: users source EC Ageing Report or OECD long-term projections for S2
- **Single contingent-liability shock**: no granular guarantee-call or SOE exposure modelling

v0.2.0 roadmap: quarterly-frequency support, Markov-switching shock models, bundled OECD ageing-cost series, and foreign-currency pass-through.

Contact, code, paper

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